

# Core Programming

## Core Programming: Reference-1

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## Core Programming: Reference-2

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## Core Programming: Reference-3

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# Data Structure

## Data Structure: Reference-1

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## Data Structure: Reference-2

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# Algorithm

## Algorithm: Reference-1

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| 22 | Sets and Disjoint Set Union – Topics 2       |
| 25 | Graphs – Topics 3                            |
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| 38 | Convex Hull – Topics 4                       |
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| 41 | Tree Vertex Splitting                        |
| 45 | Job Sequencing with Deadlines                |
| 48 | Minimum-Cost Spanning Trees – Topics 3       |
| 49 | Optimal Storage on Tapes                     |
| 50 | Optimal Merge Patterns                       |
| 51 | Single-Source Shortest Paths                 |
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| 53 | Multistage Graphs                            |
| 54 | All Pairs Shortest Paths General Weights     |
| 55 | Optimal Binary Search Trees                  |
| 56 | String Editing                               |
| 57 | 0/1-Knapsack                                 |
| 58 | Reliability Design                           |
| 59 | The Traveling Salesperson Problem            |
| 60 | Flow Shop Scheduling                         |
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| 65  | Bi-connected Components and DFS                        |
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| 67  | The 8-Queens Problem                                   |
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| 71  | Knapsack Problem                                       |
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| 79  | 0/1 Knapsack Problem – Topics 2                        |
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| 86  | Modular Arithmetic                                     |
| 87  | Even Faster Evaluation and Interpolation               |
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| 94  | Oracles and Adversary Arguments – Topics 4             |
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| 104 | Cook's Theorem                                         |
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| 125 | Polynomial Time Approximation Schemes – Topics 2       |
| 128 | Fully Polynomial Time Approximation Schemes – Topics 3 |
| 129 | Probabilistically Good Algorithms                      |
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| 146 | Sorting – Topics 4                |
| 148 | Graph Problems – Topics 2         |
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| 151 | Lower Bounds – Topics 2           |
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| 159 | Fundamental Algorithms – Topics 4 |
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| 167 | Sorting – Topics 2                |
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| 186 | Sorting – Topics 2                |
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| 04 | A Stored Program Model                                  |
| 05 | Abstractions of the RAM                                 |
| 06 | A Primitive Model of Computation: The Turing Machine    |
| 07 | Relationship Between the Turing Machine and RAM Models  |
| 08 | Pidgin ALGOL – A High-Level Language                    |
|    | <b>Design of Efficient Algorithms</b>                   |
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| 10 | Set Representations                                     |
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| 17 | Epilogue                                                |
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| 19 | Radix Sorting                                           |
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| 21 | Heapsort – an $O(n \log n)$ Comparison Sort             |
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| 23 | Order Statistics                                        |
| 24 | Expected Time for Order Statistics                      |
|    | <b>Data Structures for Set Manipulation Problems</b>    |
| 25 | Fundamental Operations on Sets                          |
| 26 | Hashing                                                 |
| 27 | Binary Search                                           |
| 28 | Binary Search Trees                                     |
| 29 | Optimal Binary Search Trees                             |
| 30 | A Simple Disjoint-Set Union Algorithm                   |
| 31 | Tree Structures for the UNION-FIND Problem              |
| 32 | Applications and Extensions of the UNION-FIND Algorithm |
| 33 | Balanced Tree Schemes                                   |
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| 39 | Depth-First Search                                      |
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| 42 | Strong Connectivity                                     |
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|    | <b>Matrix Multiplication and Related Operations</b>                   |
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| 52 | LUP Decomposition of Matrices                                         |
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| 54 | Boolean Matrix Multiplication                                         |
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| 55 | The Discrete Fourier Transform and Its Inverse                        |
| 56 | The Fast Fourier Transform Algorithm                                  |
| 57 | The FFT Using Bit Operations                                          |
| 58 | Products of Polynomials                                               |
| 59 | The Schonhage-Strassen Integer-multiplication Algorithm               |
|    | <b>Integer and Polynomial Arithmetic</b>                              |
| 60 | The Similarity Between Integers and Polynomials                       |
| 61 | Integer Multiplication and Division                                   |
| 62 | Polynomial Multiplication and Division                                |
| 63 | Modular Arithmetic                                                    |
| 64 | Modular Polynomial Arithmetic and Polynomial Evaluation               |
| 65 | Chinese Remaindering                                                  |
| 66 | Chinese Remaindering and Interpolation of Polynomials                 |
| 67 | Greatest Common Divisors and Euclid's Algorithm                       |
| 68 | An Asymptotically Fast Algorithm for Polynomial GCD's                 |
| 69 | Integer GCD's                                                         |
| 70 | Chinese Remaindering Revisited                                        |
| 71 | Sparse Polynomials                                                    |
|    | <b>Pattern-Matching Algorithms</b>                                    |
| 72 | Finite Automata and Regular Expressions                               |
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| 75 | Tow –way Deterministic Pushdown Automata                              |
| 76 | Position Trees and Substring Identifiers                              |
|    | <b>NP-Complete Problems</b>                                           |
| 77 | Nondeterministic Turing Machines                                      |
| 78 | The Classes P and NP                                                  |
| 79 | Languages and Problems                                                |
| 80 | NP-Completeness of the Satisfiability Problem                         |
| 81 | Additional NP-Complete Problems                                       |
| 82 | Polynomial-Space-Bounded Problems                                     |
|    | <b>Some Provably Intractable Problems</b>                             |
| 83 | Complexity Hierarchies                                                |
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|    | <b>Lower Bounds on Numbers of Arithmetic Operations</b>               |
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| 88 | Straight-Line Code Revisited                                          |
| 89 | A Matrix Formulation of Problems                                      |
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# Software Engineering

## Software Engineering: Reference-1

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| 17 | Computer Systems Engineering                                         |
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| 33 | A Comparison of Different Life Cycle Models – Topics 1               |
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| 28 | Project Planning – Topics 2                                          |
| 30 | Metrics for Project Size Estimation – Topics 2                       |
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# Database Management System

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| 389 | Extended Concurrency Control Protocols – Topics 3                         |
| 395 | Replication with Weak Degrees of Consistency – Topics 6                   |
| 397 | Coordinator Selection – Topics 2                                          |
| 402 | Consensus in Distributed Systems – Topics 5                               |
|     | Summary, Exercises                                                        |
|     | <b>Advanced Topics</b>                                                    |
|     | <b>Advanced Indexing Techniques</b>                                       |
| 403 | Bloom Filter                                                              |
| 408 | Log-Structured Merge Tree and Variants – Topics 5                         |
| 411 | Bitmap Indices – Topics 3                                                 |
| 413 | Indexing of Spatial Data – Topics 2                                       |
| 416 | Hash Indices – Topics 3                                                   |
|     | Summary, Exercises                                                        |
|     | <b>Advanced Application Development</b>                                   |
| 425 | Performance Tuning – Topics 9                                             |
| 428 | Performance Benchmarks – Topics 3                                         |
| 430 | Other Issues in Application Development – Topics 2                        |
| 433 | Standardization – Topics 3                                                |
| 436 | Distributed Directory Systems – Topics 3                                  |
|     | Summary, Exercises                                                        |
|     | <b>Block-chain Databases</b>                                              |
| 437 | Overview                                                                  |
| 438 | Block-chain Properties                                                    |
| 441 | Achieving Blockchain Properties via Cryptographic Has Function – Topics 3 |
| 444 | Consensus – Topics 3                                                      |
| 446 | Data Management in a Block-chain – Topics 2                               |
| 450 | Smart Contracts – Topics 4                                                |
| 453 | Performance Enhancement – Topics 3                                        |
| 454 | Emerging Application                                                      |
|     | Summary, Exercises                                                        |
|     | <b>Appendix: Detailed University Schema</b>                               |
|     | <b>Online Chapters</b>                                                    |
|     | Formal Relational Query Languages                                         |
|     | Advanced Relational Database Design                                       |
|     | Object-Based Databases                                                    |
|     | XML                                                                       |
|     | Information Retrieval                                                     |
|     | PostgreSQL                                                                |

## Pure Mathematics – 1, 2, and 3

|    | Pure Mathematics (A-Level)                          |
|----|-----------------------------------------------------|
|    | <b>P1: Algebra</b>                                  |
| 01 | Background Algebra                                  |
| 02 | Linear Equations                                    |
| 03 | Changing the Subject of a Formula                   |
| 04 | Quadratic Equations                                 |
| 05 | Solving Quadratic Equations                         |
| 06 | Equations that cannot be Factorized                 |
| 07 | The Graphs of Quadratic Function                    |
| 08 | The Quadratic Formula                               |
| 09 | Inequalities                                        |
|    | <b>P2: Algebra</b>                                  |
| 10 | Operations with Polynomials                         |
| 11 | Solution of Polynomial Equations                    |
| 12 | The Modulus Function                                |
|    | <b>P3: Further Algebra</b>                          |
| 13 | The General Binomial Expansion                      |
| 14 | Review of Algebraic Functions                       |
| 15 | Partial Functions                                   |
| 16 | Using Partial Functions with The Binomial Expansion |
|    | <b>P1: Co-Ordinate Geometry</b>                     |
| 17 | Co-Ordinates                                        |
| 18 | Plotting, Sketching and Drawing                     |
| 19 | The Gradient of a Line                              |
| 20 | The Distance Between Two Points                     |
| 21 | The Mid-Point of a Line Joining Two Points          |
| 22 | The Equation of a Straight Line                     |
| 23 | Finding the Equation of a Line                      |
| 24 | The Intersection of Two Line                        |
| 25 | Drawing Curves                                      |
| 26 | The Intersection of A Line and A Curve              |
|    | <b>P1: Sequences and Series</b>                     |
| 27 | Definitions and Notation                            |
| 28 | Arithmetic Progressions                             |
| 29 | Geometric Progressions                              |
| 30 | Binomial Expansions                                 |
|    | <b>P1: Functions</b>                                |
| 31 | The Language of Functions                           |
| 32 | Composite Functions                                 |
| 33 | Inverse Functions                                   |
|    | <b>P1: Differentiation</b>                          |
| 34 | The gradient of a Curve                             |
| 35 | Finding the Gradient of a Curve                     |
| 36 | Finding the Gradient from First Principles          |
| 37 | Differentiating by Using Standard Results           |
| 38 | Using Differentiation                               |
| 39 | Tangents and Normals                                |
| 40 | Maximum and Minimum Points                          |
| 41 | Increasing and Decreasing Functions                 |
| 42 | Points of Inflection                                |
| 43 | The Second Derivative                               |
| 44 | Applications                                        |

|    |                                                             |
|----|-------------------------------------------------------------|
| 45 | The Chain Rule                                              |
|    | <b>P2: Differentiation</b>                                  |
| 46 | The Product Rule                                            |
| 47 | The Quotient Rule                                           |
| 48 | Differentiating Natural Logarithms and Exponentials         |
| 49 | Differentiating Trigonometrical Functions                   |
| 50 | Differentiating Functions Defined Implicitly                |
| 51 | Parametric Equations                                        |
| 52 | Parametric Differentiation                                  |
|    | <b>P3: Differential Equations</b>                           |
| 53 | Forming Differential Equations from Rates of Change         |
| 54 | Solving Differential Equations                              |
|    | <b>P1: Integration</b>                                      |
| 55 | Reversing Differentiation                                   |
| 56 | Finding the Area Under A Curve                              |
| 57 | Area as The Limit of A Sum                                  |
| 58 | Areas Below the X Axis                                      |
| 59 | The Area Between Two Curves                                 |
| 60 | The Area Between A Curve and the Y Axis                     |
| 61 | The Reverse Chain Rule                                      |
| 62 | Improper Integrals                                          |
| 63 | Finding Volumes by Integration                              |
|    | <b>P2: Integration</b>                                      |
| 64 | Integrals Involving The Exponential Function                |
| 65 | Integrals Involving The Natural Logarithm Function          |
| 66 | Integrals Involving Trigonometrical Functions               |
| 67 | Numerical Integration                                       |
|    | <b>P3: Further Integration</b>                              |
| 68 | Integration by Substitution                                 |
| 69 | Integrals Involving Exponentials and Natural Logarithms     |
| 70 | Integrals Involving Trigonometrical Function                |
| 71 | The Use of Partial Fractions in Integration                 |
| 72 | Integration by Parts                                        |
| 73 | General Integration                                         |
|    | <b>P1: Trigonometry</b>                                     |
| 74 | Trigonometry Background                                     |
| 75 | Trigonometrical Functions                                   |
| 76 | Trigonometrical Functions for Angles of any Size            |
| 77 | The sine and cosine Graphs                                  |
| 78 | The tangent graph                                           |
| 79 | Solving Equations using Graphs of Trigonometrical Functions |
| 80 | Circular Measure                                            |
| 81 | The Length of An ARC of A Circle                            |
| 82 | The Area of A Sector of A Circle                            |
| 83 | Other Trigonometrical Functions                             |
|    | <b>P2: Trigonometry</b>                                     |
| 84 | Reciprocal Trigonometrical Functions                        |
| 85 | Compound-Angle Formulae                                     |
| 86 | Double-Angle Formulae                                       |
| 87 | The Forms $r\cos$ , $r\sin$                                 |
| 88 | The General Solutions of Trigonometrical Equations          |
|    | <b>P1: Vectors</b>                                          |
| 89 | Vectors in Two Dimensions                                   |
| 90 | Vectors in Three Dimensions                                 |
| 91 | Vectors Calculations                                        |

|     |                                                   |
|-----|---------------------------------------------------|
| 92  | The Angle Between Two Vectors                     |
|     | <b>P3: Vectors</b>                                |
| 93  | The Vector Equation of a Line                     |
| 94  | The Intersection of Two Lines                     |
| 95  | The Angle Between Two Lines                       |
| 96  | The Perpendicular Distance from A Point to a Line |
| 97  | The Vector Equation of a Plane                    |
| 98  | The Intersection of A Line and A Plane            |
| 99  | The Distance of A Point from A Plane              |
| 100 | The Angle Between A Line and A Plane              |
| 101 | The Intersection of Two Planes                    |
|     | <b>P2: Logarithms and Exponentials</b>            |
| 102 | Logarithms                                        |
| 103 | Exponential Functions                             |
| 104 | Modelling Curves                                  |
| 105 | The Natural Logarithm Functions                   |
| 106 | The Exponential Function                          |
|     | <b>P2: Numerical Solution of Equations</b>        |
| 107 | Interval Estimation – Change-of-Sign Methods      |
| 108 | Fixed-Point Iteration                             |
|     | <b>P3: Complex Numbers</b>                        |
| 109 | The Growth of the Number System                   |
| 110 | Working with Complex Numbers                      |
| 111 | Representing Complex Numbers Geometrically        |
| 112 | Sets of Points in An Argand Diagram               |
| 113 | The Modulus-Argument form of Complex Numbers      |
| 114 | Sets of Points Using The Polar Form               |
| 115 | Working with Complex Numbers in Polar Form        |
| 116 | Complex Exponents                                 |
| 117 | Complex Numbers and Equations                     |

# Discrete Mathematics

## Discrete Mathematics: Reference-1

|    | Discrete Mathematics – Schaum's Outlines                      |
|----|---------------------------------------------------------------|
|    | <b>Set Theory</b>                                             |
| 01 | Introduction                                                  |
| 02 | Sets and Elements, Subsets                                    |
| 03 | Ven Diagrams                                                  |
| 04 | Algebra of Sets, Duality                                      |
| 05 | Finite Sets, Counting Principle                               |
| 06 | Classes of Sets, Power Sets, Partitions                       |
| 07 | Mathematical Introduction                                     |
|    | Solved Problems                                               |
|    | Supplementary Problems                                        |
|    | <b>Relations</b>                                              |
| 08 | Introduction                                                  |
| 09 | Product Sets                                                  |
| 10 | Relations                                                     |
| 11 | Pictorial Representatives of Relations                        |
| 12 | Composition of Relations                                      |
| 13 | Types of Relations                                            |
| 14 | Closure Properties                                            |
| 15 | Equivalence Relations                                         |
| 16 | Partial Ordering Relations                                    |
|    | Solved Problems                                               |
|    | Supplementary Problems                                        |
|    | <b>Functions and Algorithms</b>                               |
| 17 | Introduction                                                  |
| 18 | Functions                                                     |
| 19 | One-to-One, Onto, and Invertible Functions                    |
| 20 | Mathematical Functions, Exponential and Logarithmic Functions |
| 21 | Sequences, Indexed Classes of Sets                            |
| 22 | Recursively Defined Functions                                 |
| 23 | Cardinality                                                   |
| 24 | Algorithms and Functions                                      |
| 25 | Complexity of Algorithms                                      |
|    | Solved Problems                                               |
|    | Supplementary Problems                                        |
|    | <b>Logic and Propositional Calculus</b>                       |
| 26 | Introduction                                                  |
| 27 | Propositions and Compound Statements                          |
| 28 | Basic Logical Operations                                      |
| 29 | Propositions and truth Tables                                 |
| 30 | Tautologies and Contradictions                                |
| 31 | Logical Equivalence                                           |
| 32 | Algebra of Propositions                                       |
| 33 | Conditional and Bi-conditional Statements                     |
| 34 | Arguments                                                     |
| 35 | Propositional Functions, Quantifiers                          |
| 36 | Negation of Quantified Statements                             |
|    | Solved Problems                                               |
|    | Supplementary Problems                                        |
|    | <b>Techniques of Counting</b>                                 |

|    |                                                              |
|----|--------------------------------------------------------------|
| 37 | Introduction                                                 |
| 38 | Basic Counting Principles                                    |
| 39 | Mathematical Functions                                       |
| 40 | Permutations                                                 |
| 41 | Combinations                                                 |
| 42 | The Pigeonhole Principle                                     |
| 43 | The Inclusion-Exclusion Principle                            |
| 44 | Tree Diagrams                                                |
|    | Solved Problems                                              |
|    | Supplementary Problems                                       |
|    | <b>Advanced Counting Techniques, Recursion</b>               |
| 45 | Introduction                                                 |
| 46 | Combinations with Repetitions                                |
| 47 | Ordered and Unordered Partitions                             |
| 48 | Inclusion – Exclusion Principle Revisited                    |
| 49 | Pigeonhole Principle Revisited                               |
| 50 | Recurrence Relations                                         |
| 51 | Linear Recurrence Relations with Constant Coefficients       |
| 52 | Solving Second-Order Homogeneous Linear Recurrence Relations |
| 53 | Solving General Homogeneous Linear Recurrence Relations      |
|    | Solved Problems                                              |
|    | Supplementary Problems                                       |
|    | <b>Probability</b>                                           |
| 54 | Introduction                                                 |
| 55 | Sample Space and Events                                      |
| 56 | Finite Probability Spaces                                    |
| 57 | Conditional Probability                                      |
| 58 | Independent Events                                           |
| 59 | Independent Repeated Trials, Binomial Distribution           |
| 60 | Random Variables                                             |
| 61 | Chebyshev's Inequality, Law of Large Numbers                 |
|    | Solved Problems                                              |
|    | Supplementary Problems                                       |
|    | <b>Graph Theory</b>                                          |
| 62 | Introduction, Data Structures                                |
| 63 | Graphs and Multigraphs                                       |
| 64 | Subgraphs, Isomorphic and Homeomorphic Graphs                |
| 65 | Paths, Connectivity                                          |
| 66 | Traversable and Eulerian Graphs, Bridges of Königsberg       |
| 67 | Labeled and Weighted Graphs                                  |
| 68 | Complete, Regular, and Bipartite Graphs                      |
| 69 | Tree Graphs                                                  |
| 70 | Planar Graphs                                                |
| 71 | Graph Colorings                                              |
| 72 | Representing Graphs in Computer Memory                       |
| 73 | Graph Algorithms                                             |
| 74 | Traveling-Salesman Problem                                   |
|    | Solved Problems                                              |
|    | Supplementary Problems                                       |
|    | <b>Directed Graphs</b>                                       |
| 75 | Introduction                                                 |
| 76 | Directed Graphs                                              |
| 77 | Basic Definitions                                            |
| 78 | Rooted Trees                                                 |
| 79 | Sequential Representation of Directed Graphs                 |

|     |                                                          |
|-----|----------------------------------------------------------|
| 80  | Warshall's Algorithm, Shortest Paths                     |
| 81  | Linked Representation of Directed Graphs                 |
| 82  | Graph Algorithms: Depth-First and Breadth-First Searches |
| 83  | Directed Cycle-Free Graphs, Topological Sort             |
| 84  | Pruning Algorithm for Shortest Path                      |
|     | Solved Problems                                          |
|     | Supplementary Problems                                   |
|     | <b>Binary Trees</b>                                      |
| 85  | Introduction                                             |
| 86  | Binary Trees                                             |
| 87  | Complete and Extended Binary Trees                       |
| 88  | Representing Binary Trees in Memory                      |
| 89  | Traversing Binary Trees                                  |
| 90  | Binary Search Trees                                      |
| 91  | Priority Queues, Heaps                                   |
| 92  | Path Lengths, Huffman's Algorithm                        |
| 93  | General (Ordered Rooted) Trees Revisited                 |
|     | Solved Problems                                          |
|     | Supplementary Problems                                   |
|     | <b>Properties of the Integers</b>                        |
| 94  | Introduction                                             |
| 95  | Order and Inequalities, Absolute Value                   |
| 96  | Mathematical Induction                                   |
| 97  | Division Algorithm                                       |
| 98  | Divisibility, Primes                                     |
| 99  | Greatest Common Divisor, Euclidean Algorithm             |
| 100 | Fundamental theorem of Arithmetic                        |
| 101 | Congruence Relation                                      |
| 102 | Congruence Equations                                     |
|     | Solved Problems                                          |
|     | Supplementary Problems                                   |
|     | <b>Languages, Automata, Grammars</b>                     |
| 103 | Introduction                                             |
| 104 | Alphabet, Words, Free Semigroup                          |
| 105 | Languages                                                |
| 106 | Regular Expressions, Regular Languages                   |
| 107 | Finite State Automata                                    |
| 108 | Grammars                                                 |
|     | Solved Problems                                          |
|     | Supplementary Problems                                   |
|     | <b>Finite State Machines and Turning Machines</b>        |
| 109 | Introduction                                             |
| 110 | Finite State Machines                                    |
| 111 | Godel Numbers                                            |
| 112 | Turning Machines                                         |
| 113 | Computable Functions                                     |
|     | Solved Problems                                          |
|     | Supplementary Problems                                   |
|     | <b>Ordered Sets and Lattices</b>                         |
| 114 | Introduction                                             |
| 115 | Ordered Sets                                             |
| 116 | Hasse Diagrams of Partially Ordered Sets                 |
| 117 | Consistent Enumeration                                   |
| 118 | Supremum and Infimum                                     |
| 119 | Isomorphic (Similar) Ordered Sets                        |

|     |                                                            |
|-----|------------------------------------------------------------|
| 120 | Well-Ordered Sets                                          |
| 121 | Lattices                                                   |
| 122 | Bounded Lattices                                           |
| 123 | Distributive Lattices                                      |
| 124 | Complements, Complemented Lattices                         |
|     | Solved Problems                                            |
|     | Supplementary Problems                                     |
|     | <b>Boolean Algebra</b>                                     |
| 125 | Introduction                                               |
| 126 | Basic Definitions                                          |
| 127 | Duality                                                    |
| 128 | Basic Theorems                                             |
| 129 | Boolean Algebras as Lattices                               |
| 130 | Representation Theorem                                     |
| 131 | Sum-of-Products Form for Sets                              |
| 132 | Sum-of-Products Form for Boolean Algebras                  |
| 133 | Minimal Boolean Expressions, Prime Implicants              |
| 134 | Logic Gates and Circuits                                   |
| 135 | Truth Tables, Boolean Functions                            |
| 136 | Karnaugh Maps                                              |
|     | Solved Problems                                            |
|     | Supplementary Problems                                     |
|     | <b>Appendix: Vectors and Matrices</b>                      |
| 137 | Introduction                                               |
| 138 | Vectors                                                    |
| 139 | Matrices                                                   |
| 140 | Matrix Addition and Scalar Multiplication                  |
| 141 | Matrix Multiplication                                      |
| 142 | Transpose                                                  |
| 143 | Square Matrices                                            |
| 144 | Invertible (Nonsingular) Matrices, Inverses                |
| 145 | Determinants                                               |
| 146 | Elementary Row Operations, Gaussian Elimination (Optional) |
| 147 | Boolean (Zero-One) Matrices                                |
|     | Solved Problems                                            |
|     | Supplementary Problems                                     |
|     | <b>Appendix: Algebraic Systems</b>                         |
| 148 | Introduction                                               |
| 149 | Operations                                                 |
| 150 | Semigroups                                                 |
| 151 | Groups                                                     |
| 152 | Subgroups, Normal Subgroups, and Homomorphisms             |
| 153 | Rings, Integral Domains, and Fields                        |
| 154 | Polynomials Over a Field                                   |
|     | Solved Problems                                            |
|     | Supplementary Problems                                     |

## Discrete Mathematics: Reference-2

| <b>Discrete Mathematics and Its Applications</b> |                                                           |
|--------------------------------------------------|-----------------------------------------------------------|
|                                                  | <b>The Foundations: Logic and Proofs</b>                  |
| 06                                               | Propositional Logic – Topics 6                            |
| 12                                               | Applications of Propositional Logic – Topics 6            |
| 19                                               | Propositional Equivalences – Topics 7                     |
| 32                                               | Predicates and Quantifiers – Topics 13                    |
| 39                                               | Nested Quantifiers – Topics 7                             |
| 47                                               | Rules of Inference – Topics 8                             |
| 56                                               | Introduction to Proofs – Topics 9                         |
| 66                                               | Proof Methods and Strategy – Topics 10                    |
|                                                  | End-of-Chapter Material                                   |
|                                                  | <b>Basic Structures: Sets, Functions, Sequences, Sums</b> |
| 74                                               | Sets – Topics 8                                           |
| 79                                               | Set Operations – Topics 5                                 |
| 85                                               | Functions – Topics 6                                      |
| 90                                               | Sequences and Summations – Topics 5                       |
| 93                                               | Cardinality of Sets – Topics 3                            |
| 97                                               | Matrices – Topics 4                                       |
|                                                  | End-of-Chapter Material                                   |
|                                                  | <b>Algorithms</b>                                         |
| 103                                              | Algorithms – Topics 6                                     |
| 108                                              | The Growth of Functions – Topics 5                        |
| 113                                              | Complexity of Algorithms – Topics 5                       |
|                                                  | End-of-Chapter Material                                   |
|                                                  | <b>Number Theory and Cryptography</b>                     |
| 118                                              | Divisibility and Modular Arithmetic – Topics 5            |
| 122                                              | Integer Representations and Algorithms – Topics 4         |
| 130                                              | Primes and Greatest Common Divisors – Topics 8            |
| 137                                              | Solving Congruencies – Topics 7                           |
| 140                                              | Applications of Congruencies – Topics 3                   |
| 149                                              | Cryptography – Topics 9                                   |
|                                                  | End-of-Chapter Material                                   |
|                                                  | <b>Induction and Recursion</b>                            |
| 157                                              | Mathematical Induction – Topics 8                         |
| 162                                              | Strong Induction and Well-Ordering – Topics 5             |
| 167                                              | Recursive Definitions and Structural Induction – Topics 5 |
| 171                                              | Recursive Algorithms – Topics 4                           |
| 176                                              | Program Correctness – Topics 5                            |
|                                                  | End-of-Chapter Material                                   |
|                                                  | <b>Counting</b>                                           |
| 182                                              | The Basics of Counting – Topics 6                         |
| 185                                              | The Pigeonhole Principle – Topics 3                       |
| 188                                              | Permutations and Combinations – Topics 3                  |
| 191                                              | Binominal Coefficients and Identities – Topics 3          |
| 196                                              | Generalized Permutations and Combinations – Topics 5      |
| 199                                              | Generating Permutations and Combinations – Topics 3       |
|                                                  | End-of-Chapter Material                                   |
|                                                  | <b>Discrete Probability</b>                               |
| 203                                              | An Introduction to Discrete Probability – Topics 4        |
| 213                                              | Probability Theory – Topics 10                            |
| 216                                              | Bayes' Theorem – Topics 3                                 |
| 224                                              | Expected Value and Variance – Topics 8                    |

|     |                                                                   |
|-----|-------------------------------------------------------------------|
|     | End-of-Chapter Material                                           |
|     | <b>Advanced Counting Techniques</b>                               |
| 227 | Applications of Recurrence Relations – Topics 3                   |
| 230 | Solving Linear Recurrence Relations – Topics 3                    |
| 232 | Divide-and-Conquer Algorithms and Recurrence Relations – Topics 2 |
| 237 | Generating Functions – Topics 5                                   |
| 239 | Inclusion-Exclusion – Topics 2                                    |
| 244 | Applications of Inclusion-Exclusion – Topics 5                    |
|     | End-of-Chapter Material                                           |
|     | <b>Relations</b>                                                  |
| 249 | Relations and Their Properties – Topics 5                         |
| 255 | n-ary Relations and Their Applications – Topics 6                 |
| 258 | Representing Relations – Topics 3                                 |
| 263 | Closures of Relations – Topics 5                                  |
| 267 | Equivalence Relations – Topics 4                                  |
| 273 | Partial Orderings – Topics 6                                      |
|     | End-of-Chapter Material                                           |
|     | <b>Graphs</b>                                                     |
| 274 | Graphs and Graph Models – Topics 1                                |
| 281 | Graph Terminology and Special Types of Graphs – Topics 7          |
| 287 | Representing Graphs and Graph Isomorphism – Topics 6              |
| 294 | Connectivity – Topics 7                                           |
| 298 | Euler and Hamilton Paths – Topics 4                               |
| 301 | Shortest-Path Problems – Topics 3                                 |
| 303 | Planar Graphs – Topics 3                                          |
| 305 | Graph Coloring – Topics 2                                         |
|     | End-of-Chapter Material                                           |
|     | <b>Trees</b>                                                      |
| 308 | Introduction to Trees – Topics 3                                  |
| 313 | Applications of Trees – Topics 5                                  |
| 317 | Tree Traversal – Topics 4                                         |
| 322 | Spanning Trees – Topics 5                                         |
| 324 | Minimum Spanning Trees – Topics 2                                 |
|     | End-of-Chapter Material                                           |
|     | <b>Boolean Algebra</b>                                            |
| 329 | Boolean Functions – Topics 5                                      |
| 331 | Representing Boolean Functions – Topics 2                         |
| 335 | Logic Gates – Topics 4                                            |
| 339 | Minimization of Circuits – Topics 4                               |
|     | End-of-Chapter Material                                           |
|     | <b>Modeling Computation</b>                                       |
| 344 | Languages and Grammars – Topics 5                                 |
| 346 | Finite-State Machines with Output – Topics 2                      |
| 351 | Finite-State Machines with No Output – Topics 5                   |
| 356 | Language Recognition – Topics 5                                   |
| 363 | Turing Machines – Topics 7                                        |
|     | End-of-Chapter Material                                           |
|     | <b>Appendices</b>                                                 |
|     | Axioms for the Real Numbers and the Positive Integers             |
|     | Exponential and Logarithmic Functions                             |
|     | Pseudocode                                                        |